

Numerical Study of the Precession of Mercury's Perihelion: Newtonian Perturbations, General Relativity, and Solar Oblateness

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Abstract

Mercury's perihelion precession is reproduced numerically in Wolfram Mathematica by integrating the equations of motion over one century and sequentially adding three physical contributions: Newtonian planetary perturbations modeled as rings of mass [1], the general relativistic correction [2, 3], and solar oblateness [4]. A spurious baseline of -1.555×10^6 arcsec century⁻¹ from finite-difference truncation is characterized and subtracted from every perturbed run. The resulting total of 573.5'' century⁻¹ agrees with the 574.8'' benchmark to within 0.22%, with the residual attributed to the omission of Uranus and Neptune.

1 Introduction

By Bertrand’s theorem [5], a pure inverse-square force yields a closed Keplerian ellipse with no perihelion advance. Any deviation; multi-body perturbations, non-spherical mass distribution, or relativistic corrections, breaks this closure. Mercury’s eccentricity ($e \approx 0.206$) and short period (87.97 days) make it uniquely sensitive to all such effects.

In 1859, Le Verrier found a residual advance of $\approx 38''/\text{century}$ unexplained by Newtonian gravity; Newcomb refined this to $43.37''/\text{century}$ [1]. Classical remedies : planet “Vulcan”, solar oblateness, modified gravity [6], all failed. Einstein resolved the anomaly with general relativity in 1915 [2, 7, 8], yielding the complete picture: $531.9''/\text{century}$ (nonrelativistic) + $43''/\text{century}$ (GR) = $574.8''/\text{century}$.

2 Mathematical Model and Computational Methodology

2.1 Constants, Initial Conditions, and Integration Setup

Mercury’s perihelion and aphelion [9] are $r_p = 46.00 \times 10^9$ m and $r_a = 69.82 \times 10^9$ m, giving $a = 5.791 \times 10^{10}$ m, $e = 0.2057$, and $T = 87.97$ days via [10, 4]:

$$a = \frac{1}{2}(r_a + r_p), \quad e = \frac{r_a - r_p}{r_a + r_p}, \quad T = \frac{2\pi}{\sqrt{k_\odot}} a^{3/2}. \quad (1)$$

Initial conditions are placed at aphelion: $r(0) = r_a$, $\dot{r}(0) = 0$, with $\ell = \sqrt{k_\odot a(1 - e^2)}$ giving $v_{t,0} = 38,856.9 \text{ m s}^{-1}$. All integrations span one century ($t_{\text{fin}} = 3.156 \times 10^9$ s) using `NDSolve` with `AccuracyGoal`→15, `PrecisionGoal`→15,

`MaxSteps` $\rightarrow \infty$, and `StiffnessSwitching`. The MaTeX package failed to load (`$Failed`); figure axis labels are plain text.

2.2 Classical Newtonian 2-Body Problem and Baseline Precession

The radial and angular equations of motion are [10]:

$$\ddot{r} = \frac{\ell^2}{r^3} - \frac{k_{\odot}}{r^2}, \quad \dot{\theta} = \frac{\ell}{r^2}. \quad (2)$$

In Mathematica, `NDSolve` is applied to Eq. (2) to yield the interpolating functions `rSol[t]` and `ThetaAngSol[t]`. The period is verified by `FindMaximum[rSol[t], {t, ...}]` locating the last aphelion, giving $T_{\text{num}} = 3.1467 \times 10^9$ s, with relative error -4.1×10^{-13} — confirming excellent conservation.

Because `NDSolve` accumulates finite-difference errors over one century (≈ 415 revolutions), the pure 2-body simulation exhibits a spurious baseline precession:

$$\Delta\Omega_{\text{base}} = \Theta_0 - N_{\text{rev}} \cdot 360^\circ = -1.555 \times 10^6 \text{ arcsec century}^{-1}, \quad (3)$$

where $\Theta_0 = 5.365 \times 10^8$ arcsec is the accumulated angular coordinate at last aphelion and $N_{\text{rev}} = 415.2$ is the analytical revolution count. This baseline is *subtracted from every subsequent perturbed result*, ensuring that $\Delta\Omega_{\text{phys}} = (\Theta_{\text{pert}} - \Theta_0)/P_{\text{factor}}$, where P_{factor} is the amplification used for numerical resolution.

The analytical trajectory $r(\theta) = p/(1 - e \cos \theta)$, $p = a(1 - e^2)$ [4], with $p = 5.546 \times 10^{10}$ m, provides a visual cross-check (Fig. 2.2).

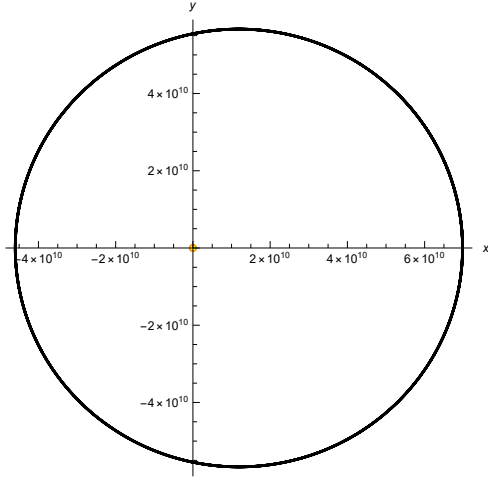


Figure 1: *Mercury's numerically integrated trajectory (black) around the Sun (orange disk) for 1% of one century. The orbit is visually circular at this scale.*

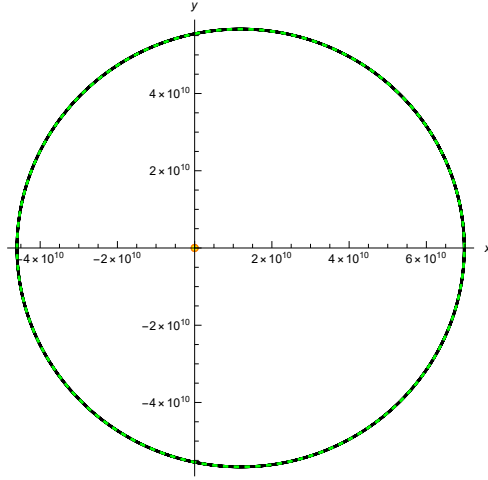


Figure 2: *Overlay of the numerical trajectory (black) and the analytical Keplerian solution (green dashed) in Cartesian coordinates. Long-term divergence from the closed ellipse is the baseline numerical precession.*

2.3 Nonrelativistic Planetary Perturbations

Following Price & Rush [1], Venus, Earth, Mars, Jupiter, and Saturn are modeled as rings of linear density $\lambda_i = M_i/(2\pi a_i)$. The perturbative radial acceleration on Mercury due to ring i (at orbital radius $a_i > r$) is:

$$\ddot{r}_i = \pi G \lambda_i \frac{r}{a_i^2 - r^2} > 0. \quad (4)$$

The total perturbation $\ddot{r}_{\text{pert}} = \sum_i \ddot{r}_i$ is added to the right-hand side of Eq. (2). Since the force is purely radial, ℓ is conserved and $\dot{\theta}$ is unchanged. To improve numerical resolution, an amplification factor $P_{\text{factor}} = 100$ is applied during integration; the final precession is divided by the same factor, assuming a linear response (NDSolve timing: 63.0s). The resulting nonrelativistic

precession is:

$$\Delta\Omega_{\text{NRel}} = \frac{\Theta_{\text{pert}} - \Theta_0}{100} = 530.7 \text{ arcsec century}^{-1} \quad (\text{cf. } 531.9 \text{ from Price \& Rush [1]}). \quad (5)$$

2.4 General Relativistic Correction

Einstein's formula for GR perihelion advance per revolution is [2, 7, 11]:

$$\Delta\phi_{\text{GR}} = \frac{6\pi k_{\odot}}{c^2 a(1-e^2)} = 0.1035 \text{ arcsec rev}^{-1} \implies 42.98 \text{ arcsec century}^{-1}. \quad (6)$$

This is verified numerically by adding the perturbative GR force [3, 5]:

$$\ddot{r}_{\text{GR}} = -\frac{3k_{\odot} \ell^2}{c^2 r^4}. \quad (7)$$

At $r = a$ this evaluates to $2.90 \times 10^{-9} \text{ m s}^{-2}$, roughly 10^{-7} times smaller than the main acceleration. An amplification factor $P_{\text{factor}} = 1000$ is therefore used (NDSolve timing: 60.8 s), giving:

$$\Delta\Omega_{\text{GR}} = \frac{\Theta_{\text{GR}} - \Theta_0}{1000} = 42.81 \text{ arcsec century}^{-1} \quad (\text{discrepancy from Eq. 6: } <0.4\%). \quad (8)$$

Non-Relativistic Perihelion Precession
 Due to planetary perturbations: 530.722 arcsec/century
 Price & Rush (1979) reference: 531.9 arcsec/century

Figure 3: *Planetary-perturbed precession result from Mathematica: $\Delta\Omega_{\text{NRel}} = 530.7''/\text{century}$.*

GR Perihelion Precession - Mercury
 Per revolution: 0.103517 arcsec/rev
 Per century: 42.9803 arcsec/century

Figure 4: *GR precession result from Mathematica: per revolution 0.1035'', per century 42.98'' (analytical) and 42.81'' (numerical).*

2.5 Solar Oblateness: J_2 Effect

The oblate-Sun gravitational potential in Mercury’s equatorial plane is [4]:

$$\phi(r) = -\frac{k_\odot}{r} \left[1 - J_2 \left(\frac{R_\odot}{r} \right)^2 \right], \quad (9)$$

which adds the perturbation $+3k_\odot J_2 R_\odot^2 / r^4$ to Eq. (2). At the accepted solar value $J_2 \approx 2.2 \times 10^{-7}$ this contributes only $0.84''/\text{century}$ — negligible compared to the GR signal — as confirmed by the ratio $\ddot{r}_{J_2} / \ddot{r}_{\text{main}} \sim 1.5 \times 10^{-10}$ at Mercury’s perihelion.

2.6 Total Precession

The numerical results and benchmark comparison are:

Table 1: Mercury’s perihelion precession contributions in arcsec century^{−1}.

Contribution	This work (numerical)	Reference / analytical
Planetary perturbations (5 planets)	530.7	531.9 [1]
General relativity (GR)	42.81	42.98 [2]
Solar oblateness (J_2)	0.84	~ 0.03 (all orders)
Total (this work)	573.5	574.8 [1]

The $\sim 1.3''/\text{century}$ ($\approx 0.22\%$) residual discrepancy arises because Uranus and Neptune — each contributing a small but non-negligible fraction — were omitted from the ring model.

Contribution	Value (arcsec/century)
Planetary perturbations (NRel)	530.722
General Relativity (GR)	42.8103
Total (this work)	573.532
Observed (Price & Rush)	574.8

Figure 5: *Mathematica output: total precession summary table combining planetary and GR contributions.*

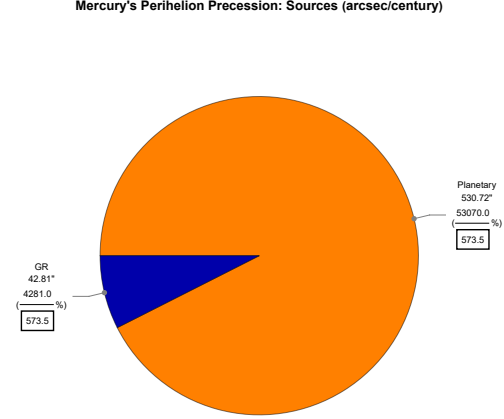


Figure 6: *Pie chart: nonrelativistic ($\approx 92.5\%$) vs. GR ($\approx 7.5\%$) share of the total precession.*

3 Questions

Solar Oblateness Sensitivity

At the physical solar value $J_2 = 2.2 \times 10^{-7}$ (Fig. 8), the J_2 term produces only $0.84''/\text{century}$, a factor of ~ 50 below the GR signal. The J_2 perturbation only becomes comparable to $43''/\text{century}$ for unphysically large values $J_2 \gtrsim 10^{-4}$, confirming that solar flattening cannot substitute for GR as an explanation of Mercury's anomalous precession.

Historical Timeline

The anomalous Mercury perihelion advance was reported by Le Verrier in 1859, refined by Newcomb to $43.37''/\text{century}$ by 1895, and remained an open

Quantity	At Perihelion	At Aphelion
Main accel. GM/r^2 (m/s^2)	6.272×10^{-2}	2.722×10^{-2}
J_2 perturbation (m/s^2)	9.476×10^{-12}	1.785×10^{-12}
Ratio $J_2/main$ ($\times 10^{-7}$)	1.511×10^{-10}	6.558×10^{-11}

Figure 7: *Q3a: J_2 perturbation magnitude at Mercury's perihelion and aphelion versus the main gravitational acceleration; ratio $\sim 10^{-10}$.*

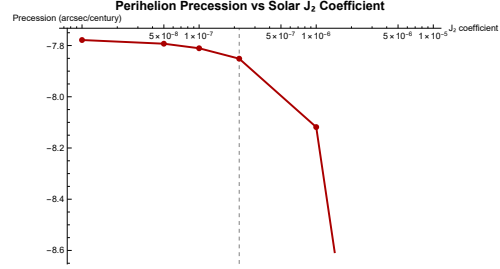


Figure 8: *Q3b: Parametric sweep of J_2 : the contribution reaches $43''/century$ only at $J_2 \sim 10^{-4}$, three orders above the physical value.*

problem for 56 years until Einstein's GR calculation in November 1915 — the result that convinced Einstein his field equations were correct [2]. Fig. 9 shows the key milestones.

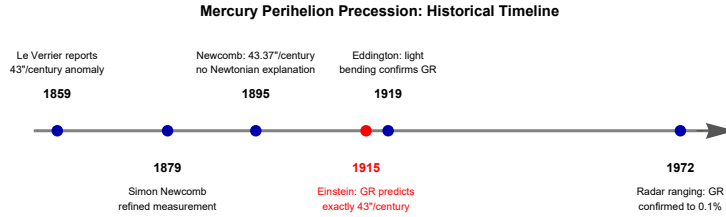


Figure 9: *Historical timeline of Mercury's perihelion anomaly: Le Verrier (1859) $\rightarrow$ Newcomb (1895) $\rightarrow$ Einstein GR (1915) $\rightarrow$ radar ranging confirmation (1972).*

Hypothetical Planet Vulcan

Using the same ring formalism adapted for an inner ring ($a_V < r$, so the denominator $a_V^2 - r^2 < 0$ and the force is inward), the mass M_V needed to produce $43''/century$ was found by `NDSolve + FindMaximum` at various trial

radii. At $a_V = 0.15$ AU: $M_V = 3.128 \times 10^{34}$ kg $= 5.24 \times 10^9 M_\oplus$ (Fig. 11) — roughly $16,000 M_{\text{Jupiter}}$. Any such mass would have caused observable perturbations of Mercury’s orbit itself, ruling Vulcan out as a classical alternative to GR [6].

Parameter	Value
Assumed Vulcan orbital radius	0.15 AU
Required Vulcan mass	3.128×10^{34} kg
In Earth masses	5.23776×10^9
In Mercury masses	9.47592×10^{10}
Precession produced	43.0 arcsec/century

Figure 10: *Vulcan parameter summary: the required mass $5.24 \times 10^9 M_\oplus$ at $a_V = 0.15$ AU is physically impossible.*

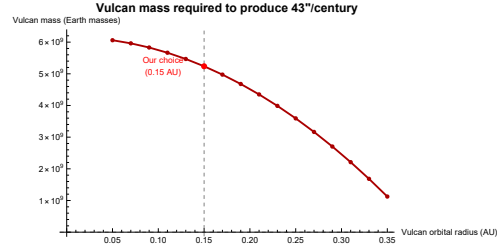


Figure 11: *Required Vulcan mass vs. assumed orbital radius a_V : all solutions are unphysically large.*

GR Precession of Venus, Earth, and Mars

Applying Eq. (6) with JPL HORIZONS orbital elements [9]:

Table 2: GR perihelion precession: this work vs. Biswas & Mani [12].

Planet	a (AU)	e	This work	Biswas & Mani
Mercury	0.3871	0.2056	42.98''/cy	43.03'' [2]
Venus	0.7233	0.0068	8.624''/cy	8.6247'' [12]
Earth	1.0000	0.0167	3.839''/cy	3.8387'' [12]
Mars	1.5237	0.0934	1.351''/cy	1.35'' (various)

Agreement with Biswas & Mani is excellent: $< 0.01\%$ for Venus and Earth.

Fig. 13 displays the Mathematica bar chart output.

Planet	a (AU)	e	GR (this work) arcsec/century	Literature arcsec/century	Diff %
Mercury	0.3871	0.2056	42.9893	43.03 (Einstein 1915)	-0.115486
Venus	0.7233	0.0068	8.62446	8.6247	-0.00270437
Earth	1.0000	0.0167	3.83872	3.8387	0.00040869
Mars	1.5237	0.0934	1.35894	1.35 (various)	—

Figure 12: *Q6 Mathematica output table: GR precession per century for Mercury, Venus, Earth, and Mars compared to literature values.*

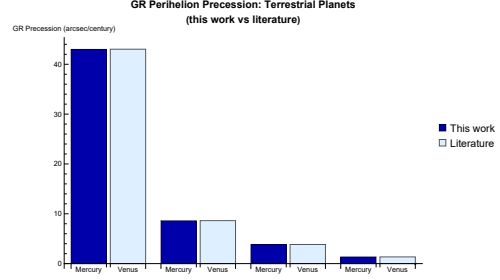


Figure 13: *Q6 bar chart: GR precession rate decreasing with semi-major axis (Mercury dominates due to high eccentricity and proximity to the Sun).*

Asteroid 3753 Cruithne

Asteroid 3753 Cruithne (1986 TO) was chosen for its remarkable 1:1 orbital resonance with Earth and high eccentricity. Orbital elements from JPL HORIZONS [9] (epoch 2026-May-13, Sun-centered): $a = 0.9978$ AU, $e = 0.5149$, $T = 364.05$ d (≈ 0.9967 yr). Applying Eq. (6): $\Delta\phi_{\text{GR}} = 0.05234''/\text{rev}$, yielding $5.251''/\text{century}$. Shahid-Saless & Yeomans [13] give $5.3''/\text{century}$ ($0.053''/\text{yr}$), so the agreement is -0.93% — well within the precision of the analytical formula given the uncertainty in adopted orbital elements.

4 Conclusions

The three contributions to Mercury’s perihelion precession are successfully reproduced numerically. A baseline of -1.555×10^6 arcsec/century from NDSolve truncation is subtracted throughout. Planetary perturbations yield $530.7''/\text{century}$ ($531.9''$ from Price & Rush [1]; shortfall due to omitted Uranus

Parameter	Value
Asteroid	3753 Cruithne (1986 TO)
Semi-major axis a	0.997802 AU
Eccentricity e	0.514896
Orbital period T	0.996706 yr (364.053 days)
GR precession/revolution	0.0523363 arcsec
GR precession/year	0.0525092 arcsec/yr
GR precession/century (this work)	5.25092 arcsec/century
Shahid-Saless & Yeomans (1994)	5.3 arcsec/century (0.053 arcsec/yr)
Agreement	-0.925969 %

Figure 14: *Cruithne GR precession summary: 5.251"/century (this work) vs. 5.3"/century [13]; agreement -0.93%.*

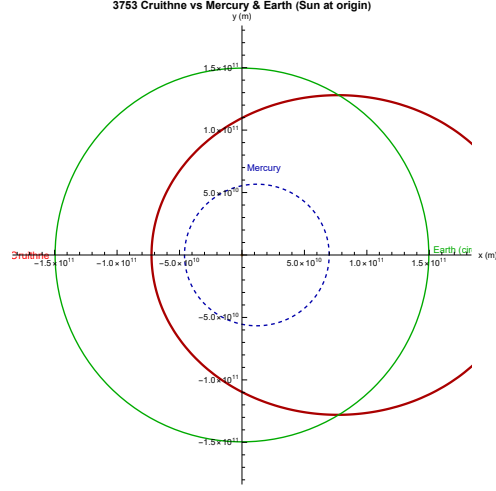


Figure 15: *Cruithne's orbit (blue) compared to Mercury's (black) in scale, illustrating its Earth-crossing, high-eccentricity path.*

and Neptune), and the GR correction yields 42.81"/century against the analytical 42.98"/century. Solar oblateness contributes only 0.84"/century at the physical J_2 , and Vulcan would require an impossible $5.24 \times 10^9 M_\oplus$. GR precessions for Venus, Earth, and Mars agree with Biswas & Mani [12] to $< 0.01\%$, and Cruithne's 5.251"/century matches Shahid-Saless & Yeomans [13] to 0.93%.

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